Successor truthfulness and particular-code knowledge

A negative answer to Alexander's Open Problem 3.2.1

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Abstract

Alexander's Open Problem 3.2.1 asks whether there can be an infinite sequence of knowing machines, each knowing the truthfulness and a code of its successor, without the ordinal knowledge assumed in his preceding impossibility theorem. We give a negative answer in his recursively named language and standard canonical semantics. The key observation is effective: particular-code knowledge for every unary modal formula makes a single enumeration of the first machine's consequences sufficient to enumerate the consequences of every later machine, uniformly. The selected indices retain their predecessors' proofs of correctness. Applying this observation to a fixed partial truth predicate produces a uniformly enumerable sequence of sound arithmetic theories, each proving uniform reflection for its successor. A theorem of Pakhomov and Walsh excludes that sequence. We give the arithmetic conversion and the comparison of proof presentations explicitly. The argument requires neither an ordinal predicate nor an internally proved totality assertion for the index-selection procedure.

1. The question and the argument

In his 2013 dissertation *The Theory of Several Knowing Machines*, Alexander studies machines that can reason about the knowledge of other machines. A machine may know that another machine is truthful, that its knowledge is recursively enumerable, or that a particular program enumerates its knowledge. These are different requirements.

Chapter 3 proves that an infinite sequence of machines cannot each know its successor's truthfulness and code when the machines also know rudimentary facts about recursive ordinals. Section 3.2 asks whether the ordinal hypothesis can be removed. The precise question is **Open Problem 3.2.1, printed page 89**; the definitions of successor truthfulness and code knowledge are in Definition 3.1.5, printed page 86 [A13]. The question occurs on page 99 of the 103-page OhioLINK PDF. It is different from the dissertation's Questions 3-1-1, 3-1-2 and 3-1-3 about well-founded hierarchies.

We answer the question by passing from modal code knowledge to arithmetic reflection. The proof has three steps.

First, a code for a machine's answers can also be requested for formulas saying that the machine endorses a code of its successor. Iterating this request allows us to reach any finite distance from the first machine. Only the first machine's enumeration is supplied to the procedure.

Second, we retain the arithmetic information needed for a reflection theorem. For each machine we form a theory consisting of a fixed weak arithmetic base and all of that machine's designated

arithmetical Π_3 consequences. Known successor truthfulness and a known particular code imply that the preceding machine proves reflection for this theory.

Third, we arrange the theories into one recursively enumerable presentation and verify that the reflection statements refer to that very presentation. The resulting sequence contradicts Pakhomov–Walsh's arithmetic non-descent theorem [PW, Theorem 3.4].

The distinction between an index that happens to be correct and an index whose correctness is proved by the predecessor is essential in the second step. The distinction between equal theorem sets and provably interchangeable proof presentations is essential in the third.

Theorem 1. No infinite successor chain

There is no family answering Alexander's Open Problem 3.2.1 affirmatively.

More explicitly, the following weaker collection of requirements is already impossible. Let L be the recursive language of arithmetic with effectively named operators $K_0, K_1, \ldots$. Let $T_0, T_1, \ldots$ be sets of L-sentences with a common standard canonical interpretation M. Assume:

1. Every consequence of every T_j is true in M.
2. Every T_j proves ordinary arithmetic PA, and the consequence set of T_0 is recursively enumerable.
3. For every $j \geq 1$ and every L-formula $\varphi(x)$ with exactly the free variable x, there is a standard number e such that
$$T_{j-1} \vdash \forall x \big(K_j \varphi(x) \leftrightarrow x \in W_{\bar{e}} \big). \tag{1}$$
4. For one fixed arithmetical Π_3 partial truth predicate $\tau(x)$, specified in Section 4, every T_j proves
$$\forall x \big(K_{j+1} \tau(x) \rightarrow \tau(x) \big). \tag{2}$$

Here and below $\bar{e}$ is the numeral for e, and W_e is the e-th recursively enumerable set in a fixed effective enumeration.

The theorem does not assume a uniform enumeration of the family. It does not even need the later consequence sets to be supplied as recursively enumerable sets: their uniform enumerability will follow from the code hypothesis. Ordinary arithmetic induction is enough; the source question assumes the stronger Peano axioms for the full modal language.

2. Canonical knowledge and particular codes

2.1. The consequence relation

We use Alexander's consequence relation for his modal language, not a stronger modal proof calculus. Write $T \vdash \theta$ when θ follows from T in that relation. Alexander defines it semantically and proves compactness and effective enumerability of validities [A13, Theorem 1.2.6]. Thus the consequences of a recursively enumerable set of axioms are recursively enumerable. Ordinary first-order arithmetic deductions remain valid deductions in this logic.

The canonical interpretation on standard arithmetic is

$$M \models K_j \varphi[s] \quad \Longleftrightarrow \quad T_j \vdash \varphi^s, \tag{3}$$

where φ^s replaces the free variables by the numerals given by s. This is the interpretation in Alexander's Definitions 1.2.15 and 3.1.4. His Lemma 1.2.12 makes numerical substitution commute with modal formation in the required way.

Self-doubting knowledge still includes truthfulness: $K_j\theta \to \theta$ holds in the intended structure. Consequently a T_j-consequence is true in that structure, by (3). The adjective *self-doubting* means that the machine need not know its own truthfulness. Our proof never adds that stronger requirement. This setting follows Alexander's modification of the earlier knowing-machine framework of Carlson, Reinhardt and Shapiro; Carlson's construction is a predecessor, not a theorem proved here [C00; A13, Sections 1.2 and 1.4].

2.2. The order of the quantifiers

For a unary formula $\varphi(x)$, introduce the unary formula

$$\mathrm{Code}_{j,\varphi}(z) := \forall x\big(K_j\varphi(x) \leftrightarrow x \in W_z\big), \tag{4}$$

with z fresh. Bound variables are renamed effectively. When this formula is itself used as the input of another code request, its free variable is renamed to the designated input variable.

Alexander's code condition is the truth in M of

$$\exists e\; K_{j-1}\,\mathrm{Code}_{j,\varphi}(e). \tag{5}$$

Because arithmetic is standard, (5) supplies an actual natural number e; clause (3) then gives precisely (1). It is not the condition

$$K_{j-1}\exists e\;\mathrm{Code}_{j,\varphi}(e). \tag{6}$$

There is no general rule extracting a particular numeral theorem from (6). The proof uses (5), which is the condition printed in Definition 3.1.5.

The free variable in (4) is z, even though j names a modal operator and φ may be modal. Thus (4), after variable renaming, is another allowed input to the full unary-formula code schema. This simple closure property drives the argument.

3. Obtaining all required indices from the first machine

Before the general construction, consider reaching T_2. To find a T_1-endorsed index for T_2's φ-answers, first ask the root enumeration for a particular code of the following T_1-column:

$$\{e : T_1 \vdash \mathrm{Code}_{2,\varphi}(\bar{e})\}. \tag{7}$$

This column is nonempty by (1). Enumerate it and take one output. That output is not just an externally correct index: it occurs because T_1 proves its correctness. Repeating the operation reaches T_3, and then every later machine.

Lemma 2. Uniform selection with predecessor endorsement

Fix an enumeration index r for the consequences of T_0. Under (1), (3) and soundness, there is a partial recursive procedure B, obtained effectively from r, such that for every $j \geq 1$ and every unary modal φ, $B(j,\ulcorner\varphi\urcorner)$ halts with an output e satisfying both

$$T_{j-1} \vdash \mathrm{Code}_{j,\varphi}(\bar{e}) \tag{8}$$

and

$$W_e = \{m : T_j \vdash \varphi(\bar{m})\}. \tag{9}$$

Proof. Use deterministic enumerations and a fixed first-output convention. Define the procedure by recursion on the positive integer j.

For $j = 1$, enumerate the consequences of T_0 using r. Search for a sentence of the exact canonical form $\mathrm{Code}_{1,\varphi}(\bar{e})$, decode its numeral, and return e.

For the step from j to $j + 1$, form

$$\chi(z) = \mathrm{Code}_{j+1,\varphi}(z).$$

Compute $d = B(j, \ulcorner \chi \urcorner)$. Enumerate W_d, and return its first output e. The recursion decreases the machine index, although the formula grows.

We prove termination, (8) and (9) together by induction on j, simultaneously for every unary modal formula. At the base, (1) promises a sentence for which the root search succeeds. That sentence is a root theorem, giving (8). Its truth in M, together with standard arithmetic and (3), gives (9).

For the induction step, χ is unary, so the induction hypothesis gives a terminating computation of d and

$$W_d = \{e : T_j \vdash \mathrm{Code}_{j+1,\varphi}(\bar{e})\}. \tag{10}$$

Condition (1) says that this set is nonempty. Its enumeration therefore produces an e. Equation (10) gives (8) for that very output. Soundness of T_j then turns the selected theorem into (9), using (3).

All formula construction, bound-variable renaming, parsing and numeral decoding are effective. A canonical spelling can be used because the permitted renamings preserve the relevant formulas. No enumeration of a later T_j is supplied to the procedure: later enumerations are obtained from indices selected by the procedure itself. This proves the lemma. □

The lemma asserts external termination on the promised inputs. It does not assert

$$T_0 \vdash \forall j \; B(j, \ulcorner \varphi \urcorner)\!\downarrow,$$

or any corresponding assertion in PA. No such theorem will be required. Notice also that the predecessor endorsement (8) was proved separately from the extensional equality (9).

4. Arithmetic theories with fixed proof presentations

The next step converts one endorsed arithmetic column into reflection. We give the details because changing an enumeration or a proof predicate without justification would leave a gap in the final application of non-descent.

4.1. Arithmetic syntax and partial truth

Initially work in the arithmetic language $L_0 = \{0, S, +, \cdot, =\}$. Designate a recursive class $\mathcal{P}_3$ of closed prenex Π_3 formulas with bounded matrices; empty blocks and effective padding are allowed. A typical member has the form

$$\forall \vec{u} \; \exists \vec{v} \; \forall \vec{w} \; \delta(\vec{u}, \vec{v}, \vec{w}),$$

where δ is bounded. Finite tuples and syntax are coded arithmetically. Primitive recursive operations used to describe the coding are abbreviations with their usual arithmetic representations, not new unproved axioms.

Fix a Π_3 partial truth predicate $\tau(z)$ for this class, taking the value false on codes outside $\mathcal{P}_3$. In particular, for each fixed $\sigma \in \mathcal{P}_3$,

$$\mathsf{I}\Sigma_1 \vdash \sigma \leftrightarrow \tau(\overline{\ulcorner \sigma \urcorner}). \tag{11}$$

For each fixed formula this disquotation is verified already in $\mathsf{I}\Sigma_1$: check its bounded matrix and then its three fixed quantifier blocks. The stronger uniform assertion about truth of all base axioms below uses PA. The construction of the predicate and those uniform equations are recalled in Lemma 3. The basic use of partial truth and witnessed proofs follows the conventions of [PW, Section 2.1].

Present $\mathsf{I}\Sigma_1$ by a fixed recursive set of axioms, each normalized to a member of $\mathcal{P}_3$. For a Σ_1 formula code c, let $J(c)$ be the normalization of its universally closed induction axiom. Include these $J(c)$ and normalizations of the fixed elementary arithmetic axioms. Denote the resulting recursive axiom predicate by $\mathrm{Base}(z)$. The normalization comes with effective logical derivations in both directions, so this is an ordinary presentation of $\mathsf{I}\Sigma_1$.

Let $E(e, z, t)$ be a primitive recursive computation-witness relation for the fixed enumeration, so $z \in W_e$ abbreviates $\exists t\, E(e, z, t)$. Define

$$U_e = \mathsf{I}\Sigma_1 + \{\sigma \in \mathcal{P}_3 : \ulcorner \sigma \urcorner \in W_e\}. \tag{12}$$

An extra-axiom line in a proof carries its accepting enumeration computation; a base-axiom line carries its base certificate. Write $\mathrm{Prf}_e(p, z)$ for the resulting primitive recursive proof-checking relation. Its existential provability predicate is Σ_1. This is a witnessed, or smooth, presentation. It also satisfies the usual derivability conditions over $\mathsf{I}\Sigma_1$.

Write $\mathrm{RFN}_3(U_e)$ for

$$\forall z \forall p \big((z \in \mathcal{P}_3 \wedge \mathrm{Prf}_e(p, z)) \rightarrow \tau(z)\big). \tag{13}$$

It expresses uniform reflection for the designated Π_3 sentences, not merely consistency. Choose a fixed-template prenex normalization $R_e \in \mathcal{P}_3$ of (13). The normalization is provably equivalent in $\mathsf{I}\Sigma_1$. A Σ_1 presentation of the proof predicate, negated in the antecedent, adds no alternation beyond Π_3.

The source theorem [PW] uses a language with exponentiation. Appendix A verifies the conservative, complexity-preserving translation between that convention and this one. We do not identify the two signatures literally.

Lemma 3. Uniform truth of the fixed base

The partial truth predicate can be chosen so that

$$\mathsf{PA} \vdash \forall z \big(\mathrm{Base}(z) \rightarrow \tau(z)\big). \tag{14}$$

Moreover, PA proves the compositional and substitution equations for bounded-level satisfaction on subformulas of prenex Σ_3 and Π_3 formulas.

Proof. Uniform evaluation of terms and bounded formulas has matching existential and universal arithmetical descriptions. Arithmetic proves their equivalence and totality. One obtains the fixed-level satisfaction predicates by quantifying over the values of each of the at most three unbounded

blocks. The existential or universal description of bounded satisfaction is chosen to match the last block. This gives the required Σ_k and Π_k predicates, $1 \le k \le 3$, without an extra alternation. Arithmetic induction on the finite syntax proves the term-substitution and compositional equations. The predicates may be combined into a single tagged satisfaction formula for the union of these classes; that combined formula need not itself be Π_3.

Let $S(c, n, a) = \mathrm{Sat}_{\Sigma_1}(c, n, a)$ express satisfaction of the Σ_1 formula coded by c, with induction variable n and remaining assignment a. One instance of full PA induction, retaining c, a as parameters, proves

$$\forall c \forall a \Big(\big[S(c, 0, a) \wedge \forall n (S(c, n, a) \to S(c, Sn, a)) \big] \to \forall n S(c, n, a) \Big). \tag{15}$$

The formula is guarded by the recursive condition that c codes a suitable Σ_1 formula. The uniform truth equations identify (15), on such codes, with truth of $J(c)$.

For completeness, the required complexity bound on $J(c)$ follows directly from writing an induction axiom for $\phi \in \Sigma_1$ as

$$\neg\phi(0) \ \vee \ \exists n \big(\phi(n) \wedge \neg\phi(Sn) \big) \ \vee \ \forall n \phi(n). \tag{16}$$

The three disjuncts are respectively in Π_1, Σ_2 and Π_2. They admit a uniform Π_3 prenex normalization, including universal closure of the parameters. This normalizes a fixed induction template, not an arbitrary formula into a falsely bounded complexity class. The truth equivalence is verified with the same fixed-level satisfaction equations. The remaining base axioms are fixed arithmetic theorems. This proves (14) and the lemma. □

Lemma 4. From column truth to reflection

Uniformly in e,

$$\mathsf{PA} \vdash \left[\forall z (z \in W_e \to \tau(z)) \to \mathrm{RFN}_3(U_e) \right]. \tag{17}$$

Proof. Work in PA, assuming the antecedent. Let p be a witnessed U_e-proof of a designated Π_3 sentence σ. Collect the finite list $\gamma_1, \ldots, \gamma_m$ of nonlogical axioms used by p, retaining their enumeration or base certificates. Every γ_i is in $\mathcal{P}_3$.

The effective deduction transformation yields a purely logical sequent derivation of

$$\neg\gamma_1, \ldots, \neg\gamma_m, \sigma. \tag{18}$$

Ordinary first-order cut elimination is formalizable in EA^+, arithmetic with total superexponentiation [PW, Section 2.1]. Therefore PA verifies the transformation to a cut-free derivation. This eliminates logical cuts; the induction axioms were retained as nonlogical premises, not eliminated by a claim about arithmetic proof strength.

The subformula property puts the formulas of the resulting derivation among term instances of subformulas of the displayed Σ_3 and Π_3 formulas. Use the tagged partial satisfaction predicate of Lemma 3. Arithmetic induction on a cut-free proof verifies soundness of its sequents: an atomic initial sequent is true, propositional rules preserve the disjunction, existential rules use the value of their witness term, and universal rules use arbitrary values for the fresh eigenvariable. Equality rules are justified by arithmetic equality and the substitution equations. The induction can be carried out uniformly over the proof code and assignments, because the satisfaction formula has fixed finite complexity.

Under the antecedent of (17), every extra γ_i is true by its enumeration certificate; every base γ_i is true by Lemma 3. Thus (18) yields $\tau(\ulcorner\sigma\urcorner)$. The choices of p and σ were arbitrary, giving (13).

Keeping the negative premises as separate sequent entries is important. We have not called an arbitrary Boolean closure of all the axioms a Π_3 formula. The proof verifies a fixed bounded-complexity base plus the specified column, and does not assert reflection for unrestricted PA. □

Lemma 5. The arithmetic information retained by a column

Suppose T contains PA, is sound in the standard canonical structure, and

$$W_e = \{m : T \vdash \tau(\bar{m})\}. \tag{19}$$

Then U_e is standard-sound and has exactly the same designated Π_3 consequences as T.

Proof. For a fixed designated Π_3 sentence σ, (11) gives

$$T \vdash \sigma \quad \Longleftrightarrow \quad T \vdash \tau(\overline{\ulcorner\sigma\urcorner}) \quad \Longleftrightarrow \quad \ulcorner\sigma\urcorner \in W_e. \tag{20}$$

Thus every such T-consequence is literally an extra axiom of U_e, with an actual enumeration witness. Conversely each extra axiom of U_e is a T-consequence; so are its $\mathsf{I}\Sigma_1$ base axioms. An ordinary arithmetic deduction from them is a valid deduction in the original consequence relation. Hence every arithmetic U_e-consequence is a T-consequence. Its axioms are true by soundness of T, so U_e is standard-sound.

This comparison is an external assertion about actual sentences and proofs. It does not assert that U_e knows an index for T, or that either theory proves the displayed equality of consequence sets. □

5. The uniform presentation and its reflection sentences

Apply Lemma 2 to the fixed formula τ, and write

$$e_j = B(j, \ulcorner\tau\urcorner), \qquad j \geq 1. \tag{21}$$

The map $j \mapsto e_j$ is recursive under the hypothesized chain, but no arithmetic proof of its totality has been obtained. We therefore use a program whose code can be compiled without waiting for this computation to finish.

Let $w(j)$ be the index of the following program:

> Compute $B(j, \ulcorner\tau\urcorner)$. If it returns e, enumerate W_e.

The fixed root index r and j are data in the program. Standard parameter compilation makes $j \mapsto w(j)$ primitive recursive independently of whether the program runs to completion. Define

$$V_j = U_{w(j)}. \tag{22}$$

These theories have a uniformly recursively enumerable axiom presentation. Extensionally, V_j and U_{e_j} have the same axioms, but that fact alone is not our justification for exchanging their reflection sentences.

Lemma 6. Pointwise provable comparison of the two presentations

For each fixed standard $j \geq 1$, $\mathsf{I}\Sigma_1$ proves equivalence of the witnessed provability predicates of V_j and U_{e_j}, preserving conclusion codes. Consequently it proves

$$\mathrm{RFN}_3(V_j) \leftrightarrow \mathrm{RFN}_3(U_{e_j}). \tag{23}$$

Proof. Lemma 2 gives an actual finite terminating computation of $B(j, \ulcorner\tau\urcorner)$ with output e_j. For this fixed j, arithmetic verifies that computation. Determinism proves uniqueness of its output.

An accepting computation of the wrapper program consists of that initial computation followed by an accepting computation for W_{e_j}. Conversely an accepting computation for W_{e_j} can be prefixed by the verified initial computation to obtain one for the wrapper. These are effective transformations of finite computation witnesses, verified in $\mathsf{I}\Sigma_1$. They prove

$$\forall z\big(z \in W_{w(j)} \leftrightarrow z \in W_{e_j}\big) \tag{24}$$

there, together with the explicit witness transformations.

Apply these transformations to the extra-axiom certificates on every line of a finite proof, leaving its arithmetic formulas, logical inferences and base certificates unchanged. Induction on the proof verifies both translations and preserves the conclusion. This proves equivalence of the provability predicates. Reflection for either presentation says that all its provable designated Π_3 sentences satisfy the same predicate τ. The provability equivalence therefore gives (23).

The verification is pointwise: a different fixed finite computation may be used for each j. It does not presuppose that $\mathsf{I}\Sigma_1$ or PA proves the uniform termination of B. □

6. The published arithmetic obstruction

Theorem 7. Pakhomov–Walsh non-descent

There is no recursively enumerable sequence $(S_n)_{n<\omega}$ of recursively enumerable extensions of EA such that S_0 is Π_3-sound and

$$S_n \vdash \mathrm{RFN}_{\Pi_3}(S_{n+1}) \qquad \text{for every } n. \tag{25}$$

This is [PW, Theorem 3.4], with that paper's arithmetized proof conventions. We invoke the theorem, rather than claiming a new arithmetic non-descent result. Its hypotheses do not require EA to prove that the sequence descends, or require the theories themselves to prove a total index-selection function.

We shall use the special case of uniformly recursively enumerable, standard-sound extensions of $\mathsf{I}\Sigma_1$. Appendix A explains the arithmetic language translation and the agreement of the reflection presentations needed to place our theories within this special case. The stronger base also avoids a hidden collection assumption when witnessing a finite proof's recursively enumerable axioms.

7. Proof of Theorem 1 and the dissertation corollary

Proof of Theorem 1. Suppose a family satisfying the theorem exists. Construct the indices e_j, the primitive recursive wrapper indices $w(j)$, and the theories V_j as above. Lemmas 2 and 5 show that all V_j are standard-sound and extend the normalized $\mathsf{I}\Sigma_1$ base.

Fix $j \geq 1$. The predecessor endorsement in Lemma 2, applied to $j+1$, gives an actual theorem

$$T_j \vdash \forall x\big(K_{j+1}\tau(x) \leftrightarrow x \in W_{\bar{e}_{j+1}}\big). \tag{26}$$

Together with the known successor truthfulness (2), arithmetic deduction inside T_j yields

$$T_j \vdash \forall x\big(x \in W_{\bar{e}_{j+1}} \to \tau(x)\big). \tag{27}$$

Since T_j contains PA, Lemma 4 gives

$$T_j \vdash \mathrm{RFN}_3(U_{e_{j+1}}). \tag{28}$$

Lemma 6, used at the particular index $j+1$, now gives

$$T_j \vdash \mathrm{RFN}_3(V_{j+1}). \tag{29}$$

The remaining issue is where this reflection is proved. Equation (29) is so far a theorem of the modal theory T_j, whereas Theorem 7 requires a proof in the arithmetic theory V_j.

Let $R_{w(j+1)}$ be the designated Π_3 normalization of the reflection sentence in (29). The normalization equivalence is provable in the base, so

$$T_j \vdash R_{w(j+1)}.$$

By (11), T_j therefore proves $\tau(\overline{\ulcorner R_{w(j+1)} \urcorner})$. By Lemma 2 this means

$$\ulcorner R_{w(j+1)} \urcorner \in W_{e_j} = W_{w(j)}. \tag{30}$$

Thus $R_{w(j+1)}$ is an **actual added axiom** of V_j, with a finite enumeration witness. Appending the base proof of its equivalence with the reflection sentence gives an actual finite proof

$$V_j \vdash \mathrm{RFN}_3(V_{j+1}). \tag{31}$$

This argument uses the external column comparison to identify a particular axiom. It does not put an unproved assertion about the whole modal theory into V_j.

The tail $V_1, V_2, \ldots$ is uniformly recursively enumerable by its compiled presentation (22). Its theories are standard-sound and satisfy (31). With the convention verified in Appendix A, it contradicts Theorem 7. This proves Theorem 1. □

Corollary 8. Negative answer to Open Problem 3.2.1

In Alexander's language and standard canonical semantics, there is no family of knowing machines in which each K_i knows the truthfulness and a particular code of K_{i+1}, as well as the Peano axioms for the full language.

Proof. The language has effective operator names. By Definition 1.2.17 and Theorem 1.2.6 of [A13], the root's consequences are recursively enumerable. The canonical clause is (3), and the Truth axiom of self-doubting knowledge makes all consequences true in the same standard structure. Knowing the Peano axioms for the full modal language includes knowing ordinary arithmetic PA. Definition 3.1.5 gives (1), as explained in Section 2.2. Knowing successor truthfulness for all formulas supplies, in particular, (2) for τ. All the hypotheses of Theorem 1 hold, which is impossible. □

8. Scope and nearby questions

8.1. How much of the code schema was used?

For the contradiction, Lemma 2 only needs to select the τ-column indices. Its recursive calls use a concrete family of formulas. At the request for the i-th machine on the way to machine j, the formula is

$$\mathrm{Code}_{i+1,\mathrm{Code}_{i+2,\ldots,\mathrm{Code}_{j,\tau}}}, \qquad 1 \le i < j, \tag{32}$$

with the indicated finite nesting and the canonical free-variable renamings. At $i = j$ the formula is τ. Each expression is finite; its modal depth grows with j. Code knowledge on this recursive family of templates suffices. We do not claim that this family is minimal.

In particular, the proof cannot be transferred merely by assertion to a code schema restricted to arithmetic formulas or to a fixed finite modal depth. No construction or impossibility result for those variants is supplied here.

8.2. An effectively named sequence is a real hypothesis

The algorithm computes the literal operator name needed at every recursive call. An arbitrary infinite path in a recursively enumerable relation need not be computable. Therefore a family over an ill-founded relation does not automatically provide the computably specified successor sequence required here.

This distinction is relevant to [A20, Theorem 79], which gives true self-referential families for some ill-founded recursively enumerable partial orders when the ordinal requirement is removed. Its existence statement does not supply a computable descending path satisfying the present hypotheses. Relabelling a noncomputable path as $0, 1, \ldots$ is not an effective change of syntax and need not preserve recursive enumerability of the root. Our conclusion is not a well-foundedness theorem for every ordinal-free endorsement relation.

8.3. What is imported and what is proved here?

The arithmetic non-descent theorem is Pakhomov–Walsh's. Alexander supplies the question, modal consequence framework and canonical semantics. The local reduction consists of the root-based selection procedure with its retained endorsements, the specified arithmetic-column conversion, and the proof-presentation comparison needed to apply non-descent. These are written out rather than hidden behind references to earlier research notes.

The proof uses ordinary published facts about arithmetized syntax, partial truth, logical cut elimination and incompleteness. It is a hand proof; it does not inherit a formal certificate from the separate repository developments concerning Questions 3-1-1 through 3-1-3. No independent human mathematical review or determination of novelty is asserted. The supplemental note records further consequences, with their proofs and exact assumptions, in the same repository.

Appendix A. Arithmetic language and reflection conventions

This appendix supplies the convention check used in Sections 4 and 6. It is not a new hypothesis on the machines.

Lemma A.1. Conservative language extension for the application of non-descent

The special case of Theorem 7 used here applies to the uniformly presented theories U_e in the language L_0, with reflection as in (13).

Proof. The language of EA in [PW] includes exponentiation. The theory $\mathsf{I}\Sigma_1$ in L_0 proves totality and uniqueness of its usual Σ_1 graph, so it admits a conservative definitional extension by that function and the elementary notational symbols in the source language. The definitions and their proof translations are fixed once for every U_e.

The complexity of the translated reflection class must also be checked. Over $\mathsf{I}\Sigma_1$, a bounded formula in the extended language has equivalent Σ_1 and Π_1 translations into L_0. To construct them, give graph values to the added-function terms; use uniqueness to obtain the universal alternative; and compose the descriptions under bounded quantification. Finite collections of evaluation witnesses are available in $\mathsf{I}\Sigma_1$. The constructions and their equivalences are uniform in bounded formula codes. These are the usual Δ_1 definitional translations.

For a prenex extended-language Π_3 sentence, choose the Π_1 version of its translated bounded matrix. Its final universal block merges with the last universal block of the given sentence. The translated sentence therefore remains Π_3. In the dual Σ_3 case, use the Σ_1 matrix translation. These fixed-class translations preserve the corresponding partial truth predicates, by induction through their three unbounded blocks. The reverse embedding of L_0 into the extended language is immediate.

Translate proofs using the graph definitions and their totality and uniqueness proofs. An induction axiom for an extended-language Σ_1 formula translates to an induction instance for its equivalent L_0 Σ_1 formula. Extra axioms are the same $\mathcal{P}_3$ sentences and retain their enumeration witnesses. Thus the proof transformations are uniform in e, preserve conclusions up to the stated translations, and are verified in the fixed base. Together with the partial-truth comparison, they transfer uniform Π_3 reflection between the two presentations. Ordinary proof predicates and witnessed proof predicates likewise agree in this base: witnesses for the finite list of enumerated axioms can be collected there. We never use an arbitrary extensionally equivalent proof predicate.

Consequently a uniformly enumerable, sound sequence of L_0 theories as in (31) would yield one of sound extensions of EA with the reflection relation of Theorem 7. The published obstruction applies. □

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[A20] Samuel Allen Alexander. Self-referential theories. *Journal of Symbolic Logic* **85**(4) (2020), 1687–1716. arXiv:2008.11535v1, 11 August 2020. The comparison in Section 8 uses Corollary 78 and Theorem 79, author-version pp. 23–25.

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[**PW**] Fedor Pakhomov and James Walsh. Reflection ranks and ordinal analysis. *Journal of Symbolic Logic* **86** (2021), 1350–1384. DOI:10.1017/jsl.2020.9. Inspected version: arXiv:1805.02095v2, 19 October 2019. The proof uses Section 2.1 for arithmetic/proof conventions and Theorem 3.4, author-version pp. 10–11, for arithmetic non-descent.